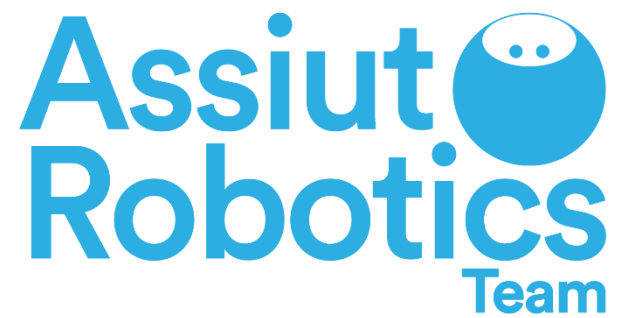


Arduino basics



Weak #02

Main objectives:

- Be familiar with viral types of sensors
- Introducing to analog and his applications
- Serial communication
- Learn how to publish your sensors reads on lcd or seven segments

Study material:

- [Simply Arduino \(Book\)](#) -> from page 70 to page 112
- [Arduino course](#) -> 8, 10, 11, 37, 38, 39, 12, 26, 16, 23
(please follow the sequence on watching video)

1st_Task:

Control LED with potentiometer, using map and constrain functions

2nd_Task:

Develop a control system to measure the temperature using any temperature sensor, you have declare the measurement via two seven segments, and while the temperature is below 30 degree green Led must be flash, but whenever the temperature exceed the 40 red led must be "blink", and between 30 and 40 degrees yellow must be flash

3rd_Task:

Distance Measurement and Display. This project uses an ultrasonic sensor to measure distance, displays the distance on the LCD, and uses push buttons and a potentiometer" control lcd brightness". the project must display the measured value in "cm" and if push button entered once the measured value should convert to "inch" also, the results should display in the "serial monitor". Also, in the same project if the measured value highest than 200 cm by using display a warning message and light a warning led.

BONUS: visit this documentation to know how to make the lcd blinking while the value is highest than 200 cm, also create a special char as the degree symbol " ° "